

Bivariate Relationships

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Relationships Between Variables



Bivariate

- + So far, we have been analyzing summary statistics that describe aspects of a single list of numbers
- + Frequently, however, we are interested in how variables behave together
 - + Measuring the relationship (correlation)
 - + Modeling the relationship (regression)



Motivating Example: Smoking and Lung Capacity

- + Suppose, for example, we wanted to investigate the relationship between cigarette smoking and lung capacity
- + We might ask a group of people about their smoking habits, and measure their lung capacities

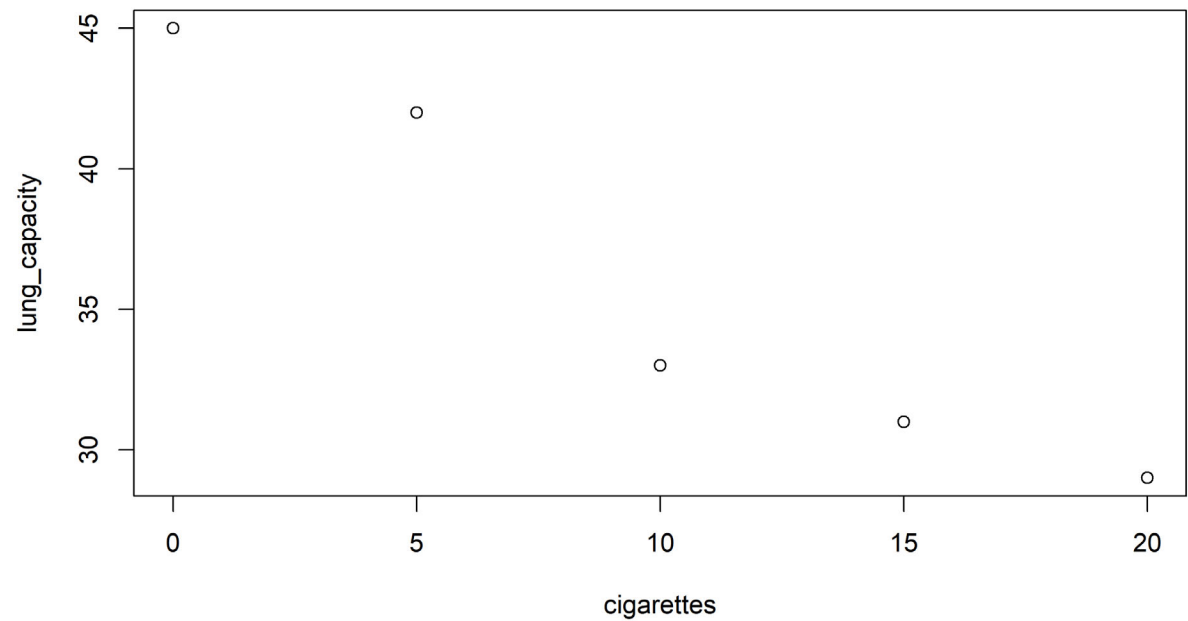
cigarettes	lung_capacity
0	45
5	42
10	33
15	31
20	29



Visualizing

- + With R or any other statistics software, we can produce a scatterplot, like this one.

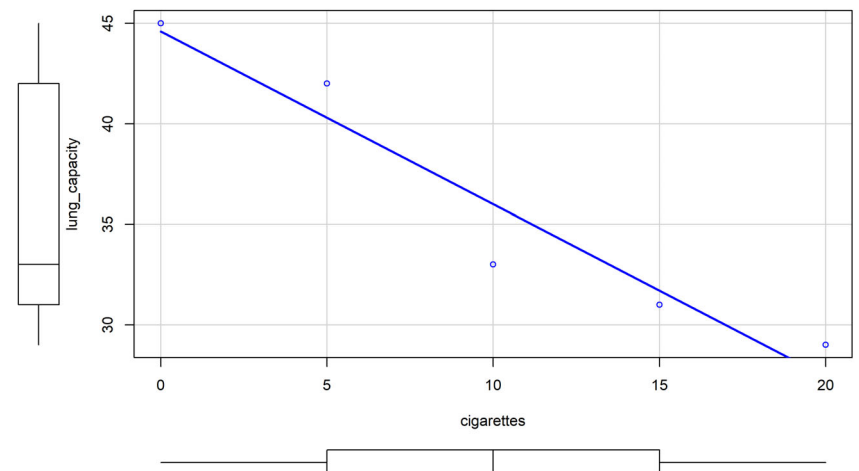
```
plot(smoking)
```



Or this one

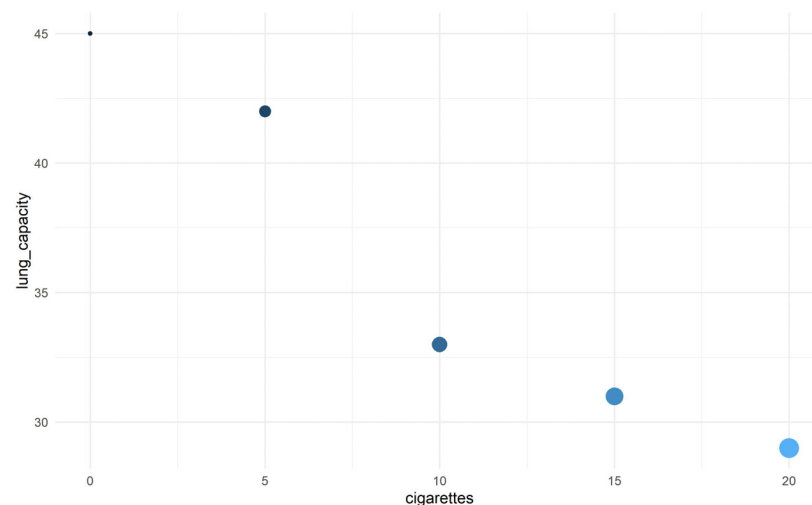
- + A scatterplot shows the relationship between two quantitative variables that are measured on the same individuals.
- + The values of one variable appear on the horizontal axis,
- + and the values of the other variable appear on the vertical axis.
- + Each individual in the data appears as the point in the plot fixed by the values of both variables for that individual.

```
library(car)  
scatterplot(lung_capacity~cigarette
```



Or this one

- + Always plot the explanatory variable,
 - + if there is one, on the horizontal axis (the x-axis) of a scatterplot
- + If there is no explanatory-response distinction,
 - + either variable can go on the horizontal axis.



```
library(ggplot2)
library(ggExtra)

# classic plot
p <- ggplot(smoking, aes(x=cigarettes,
                        y=lung_capacity,
                        color = cigarettes,
                        size=cigarettes)) +
  geom_point() + theme_minimal() +
  theme(legend.position="none")

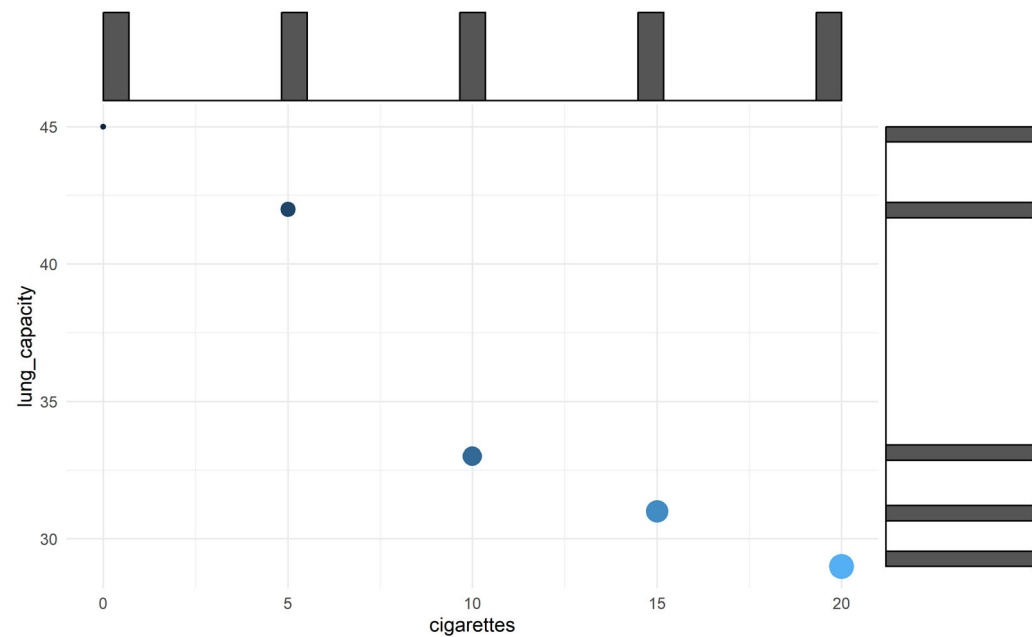
p
```



Or this one

- + We can see from the graph that as smoking goes up, lung capacity tends to go down.

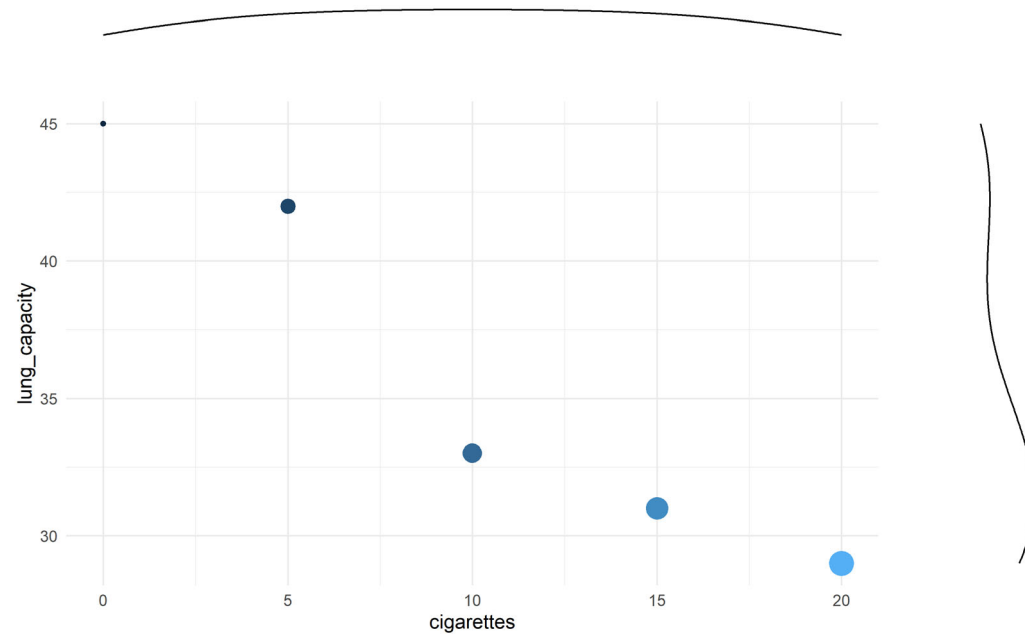
```
# with marginal histogram  
ggMarginal(p, type = "histogram")
```



Or this one

- + Here, the two variables covary in opposite directions.
- + This is a negative relationship

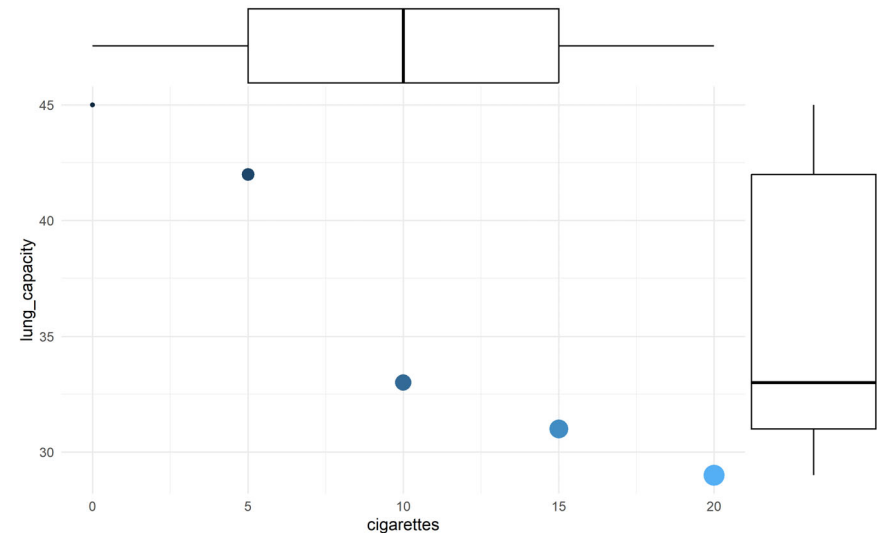
```
# marginal density  
ggMarginal(p, type="density")
```



Scatter Plots

- + In any graph of data, look for the overall pattern and for striking deviations from that pattern.
- + You can describe the overall pattern of a scatterplot by the
 - + direction (positive or negative),
 - + form, and
 - + strength of the relationship.

```
# marginal boxplot  
ggMarginal(p, type="boxplot")
```

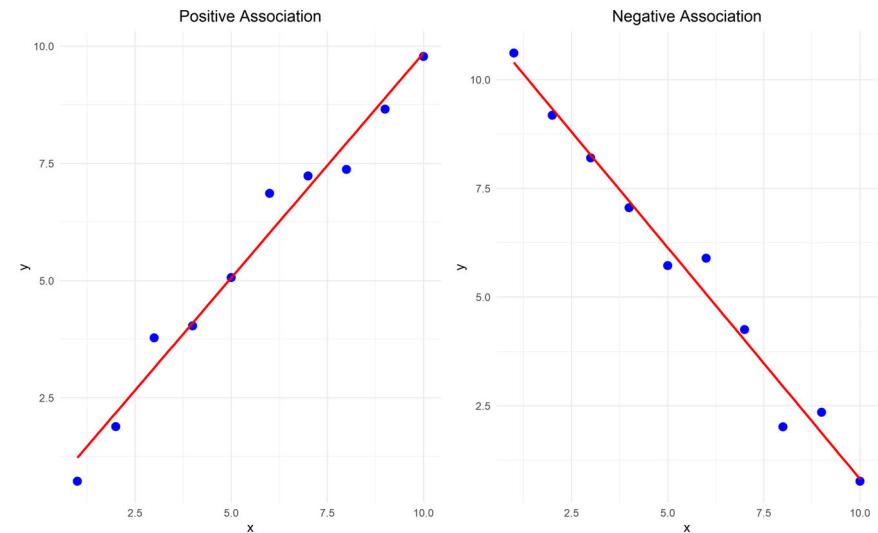


- + More examples <https://www.statmethods.net/graphs/scatterplot.html>



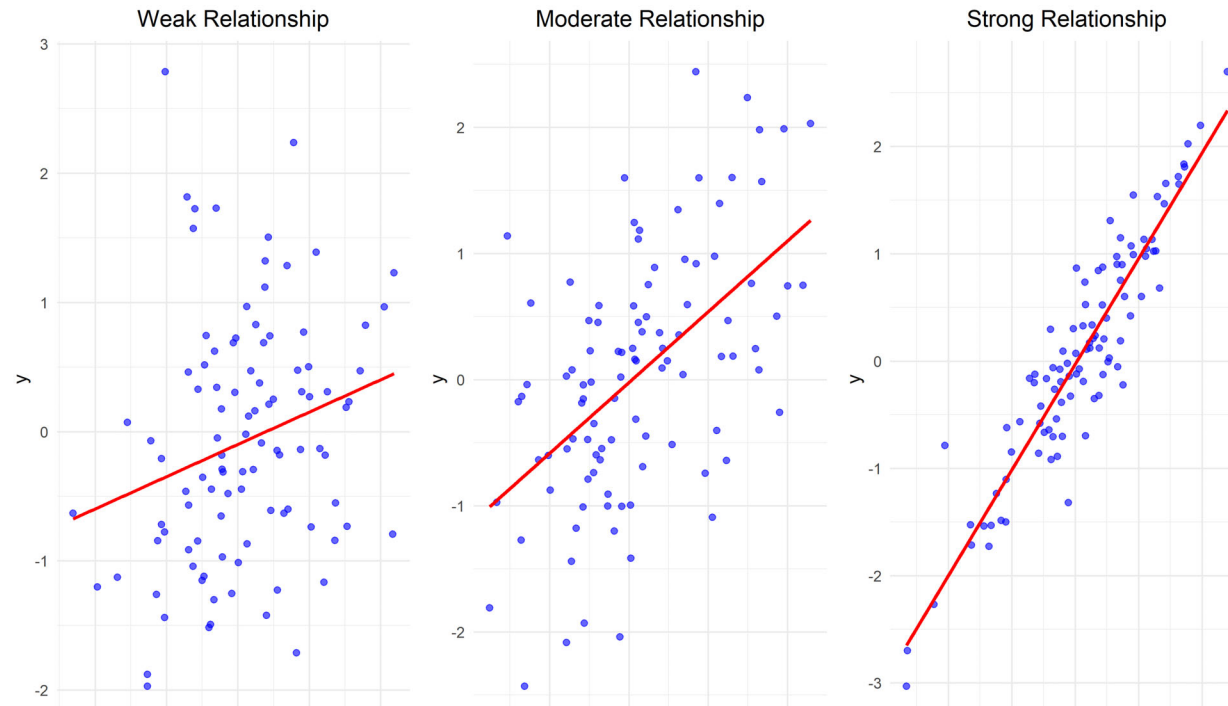
Direction of Association

- + Two variables are **positively associated** when above-average values of one tend to accompany above-average values of the other, and below-average values also tend to occur together.
- + Two variables are **negatively associated** when above-average values of one tend to accompany below-average values of the other, and vice versa.



Strength of the Relationship

- + We now examine two statistics for quantifying how variables covary:
 - + covariance and
 - + correlation.



Covariance



Covariance

- + Covariance is a measure of how much two random variables vary together.
- + When two variables covary in opposite directions, as smoking and lung capacity do,
 - + values tend to be on opposite sides of the group mean.
 - + That is, when smoking is above its group mean,
 - + lung capacity tends to be below its group mean.
- + Consequently, by averaging the product of deviation scores, we can obtain a measure of how the variables vary together.



Sample Covariance

- + Sample covariance between X and Y is an estimate of that average (cross-) product of deviation scores in the population

$$s_{x,y} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- + A more computationally convenient formula:

$$s_{x,y} = \frac{1}{n-1} \left(\sum_{i=1}^n x_i y_i - \frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n} \right)$$

- + Useful fact: the variance of a variable is its covariance with itself.

$$s^2 = s_{x,x} = \frac{1}{n-1} \sum_{i=1}^n ((x_i - \bar{x})(x_i - \bar{x}))$$



Computing Covariance

- + If we wanted to compute the covariance for smoking and lung capacity, we can save ourselves some heartache.
- + Rather than finding each deviation from the mean for x and y

```
covariance_calc=data.frame(X=smoking$cigarettes,Y=smoking$lung_capacity)

# find  $x_i - \bar{x}$ 
covariance_calc$dx=covariance_calc$X-mean(covariance_calc$X)

# find  $y_i - \bar{y}$ 
covariance_calc$dy=covariance_calc$Y-mean(covariance_calc$Y)
```

- + and then finding their product

```
#cross multiply x deviations and y deviations
covariance_calc$dx*dy=covariance_calc$dy*covariance_calc$dx

# cross multi X and Y
covariance_calc$xy=covariance_calc$Y*covariance_calc$X
```



Computing Covariance

- + Giving these values

```
kable(covariance_calc)
```

X	Y	dx	dy	dxdy	xy
0	45	-10	9	-90	0
5	42	-5	6	-30	210
10	33	0	-3	0	330
15	31	5	-5	-25	465
20	29	10	-7	-70	580

- + We'd then sum the dXdY column and get -215, and we then compute the covariance as:

$$s_{x,y} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{-215}{5-1} = -53.75$$



Alternative Computation

+ We can compute $\sum x = 50$, $\sum y = 180$, $\sum xy = 1585$, and n , and use that easier equation

$$\begin{aligned}s_{x,y} &= \frac{1}{n-1} \left(\sum x_i y_i - \frac{\sum x_i \sum y_i}{n} \right) \\&= \frac{1}{5-1} \left(1585 - \frac{50 \times 180}{5} \right) \\&= \frac{1}{4} \left(1585 - \frac{9000}{5} \right) \\&= \frac{1}{4} (1585 - 1800) \\&= \frac{1}{4} (-215) \\&= -53.75\end{aligned}$$



Covariance in R

```
cov(smoking)
```

```
##               cigarettes lung_capacity
## cigarettes      62.50      -53.75
## lung_capacity  -53.75      50.00
```

```
cov(smoking)[1,2]
```

```
## [1] -53.75
```

```
var(smoking$cigarettes)
```

```
## [1] 62.5
```



Covariance

- + Although covariance is an *extremely* important concept and is the cornerstone of many advanced methods (ANOVA, ANCOVA, SEM, regression, etc), it has some limitations:
- + it has interpretation problems just like variance
- + it isn't in a meaningful scale
- + it tells us whether a relationship is positive or negative
 - + but not much more than that.



Covariance Scaling

- + Is -53.75 a strong relationship?
 - + It depends on the scale
- + If we convert cigarettes into packs, the relationship hasn't changed.
- + but the covariance has...

```
smoking$packs=smoking$cigarettes/20  
cov(smoking$packs,smoking$lung_capacity)
```

```
## [1] -2.6875
```

- + The value is, in a sense, "polluted by the metric of the numbers."
- + Depending on the scale of the data, the absolute value of the covariance can be very large or very small



Standardizing Covariance

- + We can take the scale out of the covariance.
- + What happens if we use z-scores instead of raw deviations?
- + Remember, that z-scores are also a measure of deviations
- + This is called the (Pearson) Correlation Coefficient



Correlation Coefficient

- + The sample correlation is the sum of cross-products of z-scores divided by n-1:

$$r_{x,y} = \frac{1}{n-1} \sum_{i=1}^n (Z_{x_i} \times Z_{y_i})$$

- + The sample correlation is the sum of cross-products of z-scores divided by n:

$$\rho_{x,y} = \frac{1}{n} \sum_{i=1}^n (Z_{x_i} \times Z_{y_i})$$



Formulae for Correlation

+ We can think of a correlation coefficient as a covariance with the standard deviations factored out:

$$r_{x,y} = \frac{s_{x,y}}{s_x \times s_y}$$

+ we can think of covariance as a correlation with the standard deviations put back in.

$$s_{x,y} = r_{x,y} s_x s_y$$



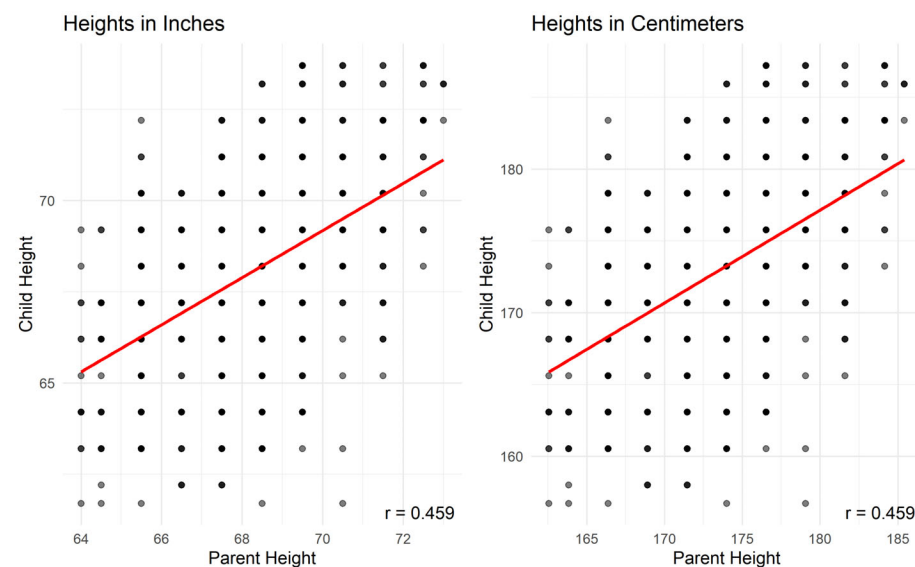
Properties of r

+ Notation

- + Sample Statistic r
- + Population Parameter ρ
 - + pronounced row, but spelt rho

+ Transformation

- + r is the same regardless of how the values are measured.
- + Height in inches and centimeters lead to the same r



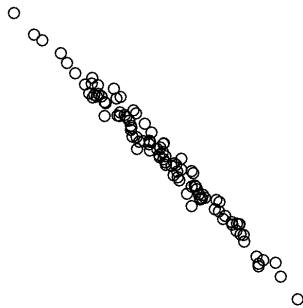
Properties of r

- + r ranges from -1 to 1 $[-1,1]$
 - + $|r|$ indicates the size of the relationship
 - + $r > 0$ indicates positive relationship $(0,1]$
 - + $r < 0$ indicates negative relationship $[-1,0)$
 - + r is symmetric
 - + $r_{x,y} = r_{y,x}$

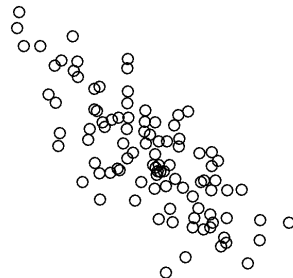


Correlation Magnitude

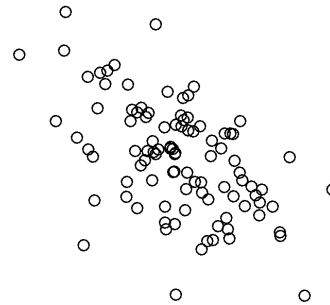
Correlation = -0.99



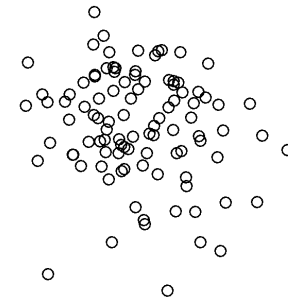
Correlation = -0.75



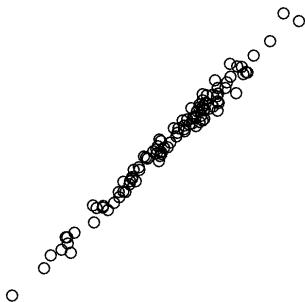
Correlation = -0.5



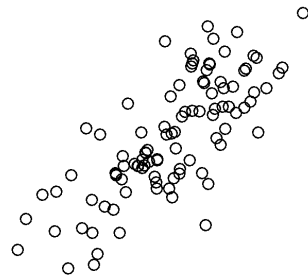
Correlation = -0.25



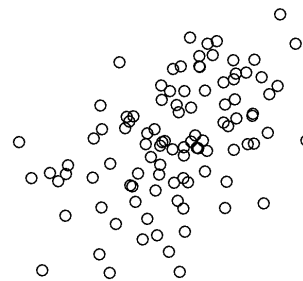
Correlation = 0.99



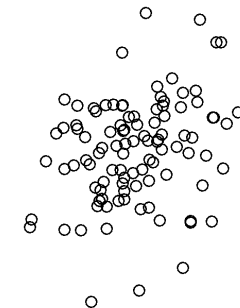
Correlation = 0.75



Correlation = 0.5



Correlation = 0.25



Correlation Interpretation

+ Perfect

- + Exactly -1 . A perfect downhill (negative) linear relationship
- + Exactly $+1$. A perfect uphill (positive) linear relationship

+ Strong

- + -0.70 . A strong downhill (negative) linear relationship
- + $+0.70$. A strong uphill (positive) linear relationship

+ Moderate

- + -0.50 . A moderate downhill (negative) relationship
- + $+0.50$. A moderate uphill (positive) relationship

+ Small

- + -0.30 . A small downhill (negative) linear relationship
- + $+0.30$. A small uphill (positive) linear relationship

- + Develop your correlation intuition

<http://guessthecorrel>



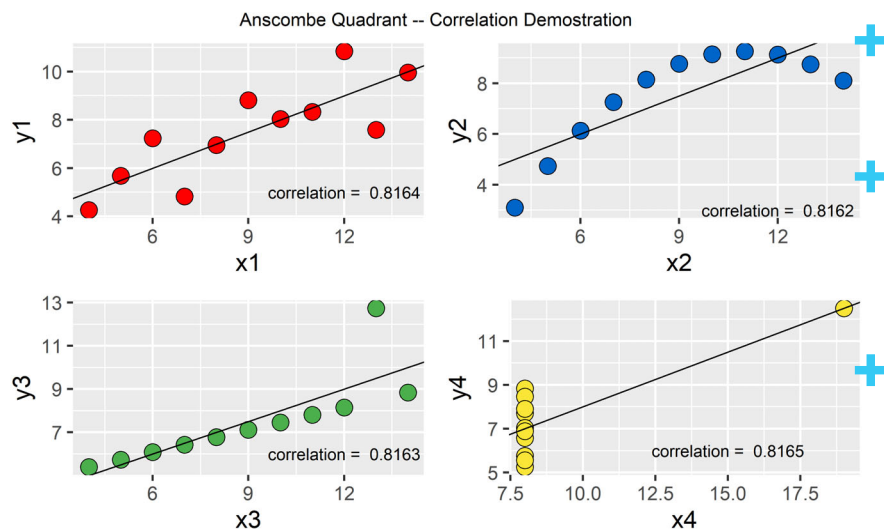
Linear Relationship

- + The correlation quantifies the linear association.
- + If your data aren't linear, then the correlation can be misleading.
- + If your data have outliers, then the correlation can be misleading.



Anscombe's Quartet

- Anscombe's Quartet comprises four datasets that have nearly identical simple descriptive statistics, yet appear very different when graphed.



These data were constructed in 1973 by the statistician Francis Anscombe (1973).

Demonstrates the importance of graphing data before analyzing it and the effect of outliers on statistical properties.

It's intended to counter the impression among statisticians that "numerical calculations are exact, but graphs are rough."



Correlation Cautions

- + Correlation requires that both variables be quantitative, so it makes sense to do the arithmetic indicated by the formula for r .
- + Correlation does not describe curved relationships between variables, no matter how strong the relationship is between them.
- + Correlation is strongly affected by a few outlying observations.
- + Correlation is not a complete summary of two-variable data.



More Correlation Cautions

- + Restriction of Range.
- + Results in r to go down artificially
 - + $r \rightarrow \rho$
- + Combined groups can hid the relationships, when groups have different relationships
 - + Simpson's Paradox



Simpson's Paradox

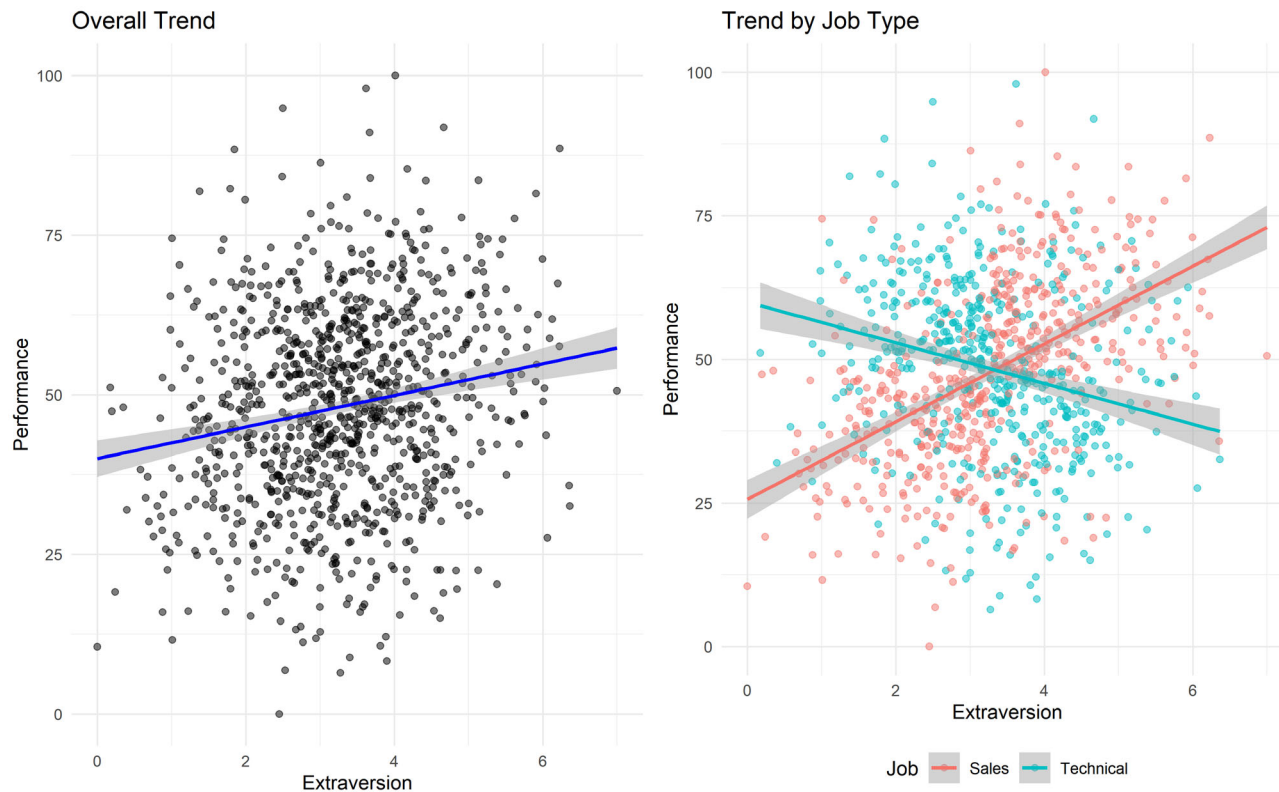
Simpson's Paradox occurs when a trend appears in different groups of data but disappears or reverses when these groups are combined.

Let's explore this through two examples:

1. Extraversion and Job Performance
2. Extraversion, Education, and Salary



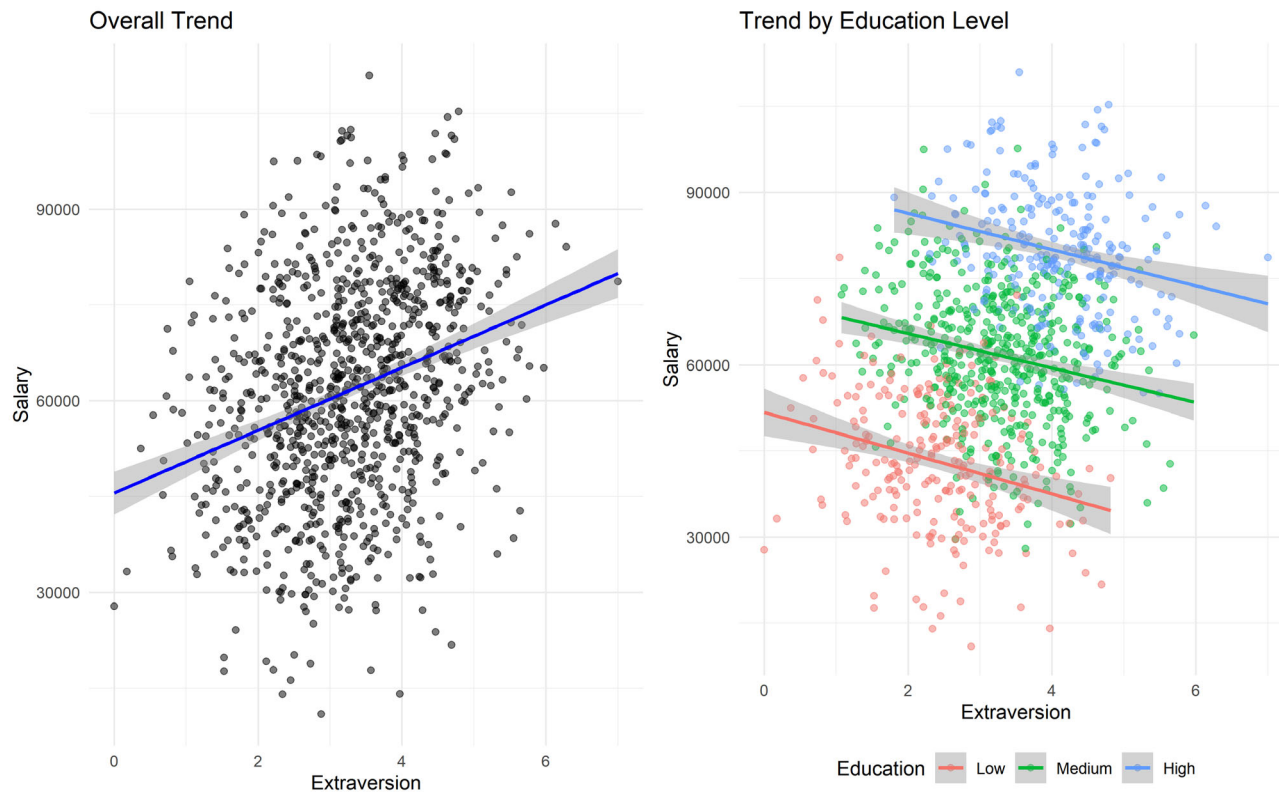
Example 1: Extraversion and Job Performance



- + Left: Overall, there's a slight positive relationship between Extraversion and Performance.
- + Right: When separated by job type, we see negative relationships within each group.
- + This illustrates Simpson's Paradox: the trend in the overall data is different from the trends in the subgroups.



Example 2: Extraversion, Education, and Salary

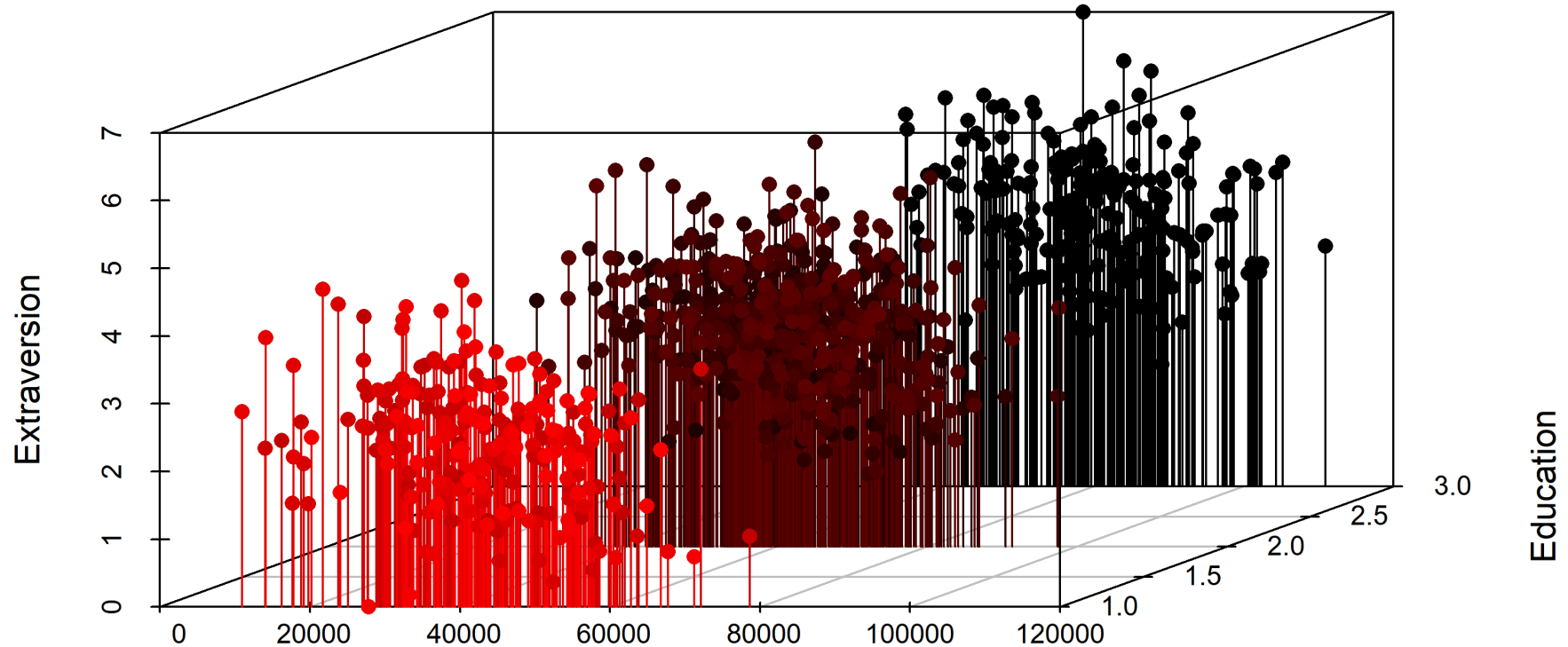


- + Left: Overall, there's a positive relationship between Extraversion and Salary.
- + Right: When separated by Education level, we see negative relationships within each group.
- + This is another example of Simpson's Paradox: the overall trend is reversed when we look at subgroups.



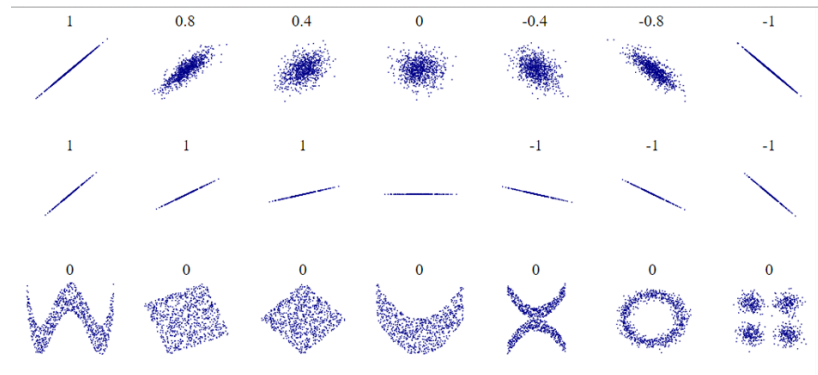
Simpson's Paradox: Illustrated

3D Scatterplot



Final Comments on the Correlation

- + The correlation coefficient is neither robust nor resistant.
 - + Not robust because strong nonlinear relationships between the two variables may not be recognized.
 - + Not resistant because it is sensitive to outlying points.
 - + It is the most sensitive summary stat we have seen thus far



Shared Variance: R^2

- + How much do those correlated variables have in common?
<http://rpsychologist.com/d3/correlation/>
- + Common variance
 - + i.e. Shared Variance
- + Coefficient of Determination
 - + % variability between the two variables that has been accounted for
 - + the remaining $1 - R^2$ of the variability is still unaccounted for



Wrapping Up...



Magnitude Code

```
# Load necessary libraries
library(ggplot2)
library(MASS)
library(gridExtra)

# Function to create correlated data and plot
create_plot <- function(correlation) {
  Sigma <- matrix(c(1, correlation, correlation, 1), 2, 2)
  data <- as.data.frame(mvrnorm(n = 100, mu = c(0, 0), Sigma = Sigma))
  colnames(data) <- c("x", "y")
  plot <- ggplot(data, aes(x = x, y = y)) +
    geom_point(shape = 1) +
    ggtitle(paste("Correlation =", correlation)) +
    theme_minimal() +
    theme(
      plot.title = element_text(size = 12, hjust = 0.5, face = "bold"),
      axis.text = element_blank(),
      axis.ticks = element_blank(),
      panel.grid = element_blank(),
      axis.title = element_blank()
    ) +
    xlim(-3, 3) + ylim(-3, 3) +
    coord_fixed(ratio = 1)
  return(plot)
}

# List of correlations
correlations <- c(-0.99, -0.75, -0.5, -0.25, 0.99, 0.75, 0.5, 0.25)

# Create list of plots
plots <- lapply(correlations, create_plot)

# Arrange plots in a grid
grid.arrange(grobs = plots, ncol = 4)
```



Methods in Psychological Research

Scale

association code

```
library(ggplot2)
library(gridExtra)

# Create data for positive association
set.seed(123) # for reproducibility
pos_data <- data.frame(
  x = 1:10,
  y = 1:10 + rnorm(10, sd = 0.5)
)

# Create data for negative association
neg_data <- data.frame(
  x = 1:10,
  y = 10:1 + rnorm(10, sd = 0.5)
)

# Plot for positive association
pos_plot <- ggplot(pos_data, aes(x, y)) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  ggtitle("Positive Association") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))

# Plot for negative association
neg_plot <- ggplot(neg_data, aes(x, y)) +
  geom_point(color = "blue", size = 3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  ggtitle("Negative Association") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))

# Combine plots
grid.arrange(pos_plot, neg_plot, ncol = 2)
```



Strength Code

```
library(ggplot2)
library(gridExtra)

set.seed(123) # for reproducibility

# Function to generate data with different strengths of relationship
generate_data <- function(n, strength) {
  x <- rnorm(n)
  y <- strength * x + rnorm(n, sd = sqrt(1 - strength^2))
  data.frame(x = x, y = y)
}

# Generate data
weak_data <- generate_data(100, 0.3)
moderate_data <- generate_data(100, 0.6)
strong_data <- generate_data(100, 0.9)

# Create plots
plot_data <- function(data, title) {
  ggplot(data, aes(x, y)) +
    geom_point(color = "blue", alpha = 0.6) +
    geom_smooth(method = "lm", se = FALSE, color = "red") +
    ggtitle(title) +
    theme_minimal() +
    theme(plot.title = element_text(hjust = 0.5))
}

weak_plot <- plot_data(weak_data, "Weak Relationship")
moderate_plot <- plot_data(moderate_data, "Moderate Relationship")
strong_plot <- plot_data(strong_data, "Strong Relationship")

# Combine plots
grid.arrange(weak_plot, moderate_plot, strong_plot, ncol = 3)
```



Weak Relationship

Moderate Relationship

Strong Relationship

Methods in Psychological Research

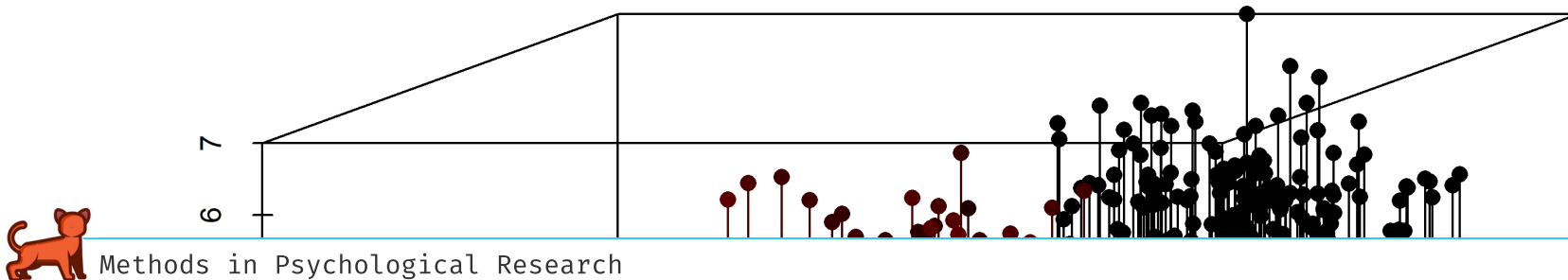
Simpsons's Code

```
library(scatterplot3d)
set.seed(123)
n = 1000
Education = rbinom(n, 2, 0.5)
Extraversion = rnorm(n) + Education
Salary = Education * 2 + rnorm(n) - Extraversion * 0.3
Salary = sample(10000:11000, 1) + rescale(Salary, to = c(0, 100000))
Extraversion = rescale(Extraversion, to = c(0, 7))
Education = factor(Education, labels = c("Low", "Medium", "High"))

data <- data.frame(Salary, Extraversion, Education)

attach(data)
s3d1 <- scatterplot3d(Salary,
  Education, Extraversion,
  pch=16, highlight.3d=TRUE,
  type="h", main="3D Scatterplot")
```

3D Scatterplot



Anscombe's Code

```
cor1 <- format(cor(anscombe$x1, anscombe$y1), digits=4)
cor2 <- format(cor(anscombe$x2, anscombe$y2), digits=4)
cor3 <- format(cor(anscombe$x3, anscombe$y3), digits=4)
cor4 <- format(cor(anscombe$x4, anscombe$y4), digits=4)

#define the OLS regression
line1 <- lm(y1 ~ x1, data=anscombe)
line2 <- lm(y2 ~ x2, data=anscombe)
line3 <- lm(y3 ~ x3, data=anscombe)
line4 <- lm(y4 ~ x4, data=anscombe)

circle.size = 5
colors = list('red', '#0066CC', '#4BB14B', '#FCE638')

#plot1
plot1 <- ggplot(anscombe, aes(x=x1, y=y1)) + geom_point(size=circle.size, pch=21, fill=colors[[1]]) +
  geom_abline(intercept=line1$coefficients[1], slope=line1$coefficients[2]) +
  annotate("text", x = 12, y = 5, label = paste("correlation = ", cor1))

#plot2
plot2 <- ggplot(anscombe, aes(x=x2, y=y2)) + geom_point(size=circle.size, pch=21, fill=colors[[2]]) +
  geom_abline(intercept=line2$coefficients[1], slope=line2$coefficients[2]) +
  annotate("text", x = 12, y = 3, label = paste("correlation = ", cor2))

#plot3
plot3 <- ggplot(anscombe, aes(x=x3, y=y3)) + geom_point(size=circle.size, pch=21, fill=colors[[3]]) +
  geom_abline(intercept=line3$coefficients[1], slope=line3$coefficients[2]) +
  annotate("text", x = 12, y = 6, label = paste("correlation = ", cor3))

#plot4
plot4 <- ggplot(anscombe, aes(x=x4, y=y4)) + geom_point(size=circle.size, pch=21, fill=colors[[4]]) +
  geom_abline(intercept=line4$coefficients[1], slope=line4$coefficients[2]) +
  annotate("text", x = 15, y = 6, label = paste("correlation = ", cor4))

grid.arrange(plot1, plot2, plot3, plot4, top='Anscombe Quadrant -- Correlation Demonstration')
```

